

CLIENT REQUIREMENTS SPECIFICATION

Kagiso Mechanical Engineering: Enterprise Network Infrastructure Upgrade

Project Reference: CMPG325/2026/066 **Client Identifier:** CLI-066 **Economic Sector:** Mechanical Engineering & Mining Support Services **Document Issue:** Revision 1.2 **Date of Issue:** 28 August 2026 **Client Organisation:** Kagiso Mechanical Engineering (Pty) Ltd, Rustenburg **Compiled By:** M. Mbodi – Student Number 40779750 **Academic Institution:** North-West University (Mahikeng Campus) **Module Code:** CMPG 325 **Academic Year:** 2026

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1. PROJECT SYNOPSIS

This technical specification captures the end-to-end networking requirements for **Kagiso Mechanical Engineering (Pty) Ltd**, a Rustenburg-based mechanical engineering concern that provides specialist services to the platinum mining belt of the North West Province. The mandate of this project is to conceptualise, simulate (via Cisco Packet Tracer), and ultimately deploy a modern, segmented, and resilient Local Area Network that replaces the organisation's legacy peer-to-peer network.

At the core of this initiative is the requirement to segment the network according to five operational departments using VLAN technology, and to carve the assigned private address space **192.168.35.0/24** into efficient, scalable subnets. An additional management subnet is set aside for server infrastructure and network appliance addressing. The solution must cater for a staff complement of approximately 52 employees distributed across administrative, engineering, production, financial, and technical support functions.

This document serves as the contractual baseline against which the design will be validated at Milestone 1 (Client Design Review – 28 August 2026), and subsequently against which implementation conformance will be measured at Milestone 2 and Final Submission.

2. KAGISO MECHANICAL ENGINEERING: ORGANISATIONAL BACKGROUND

2.1 Corporate Identity and Geographic Footprint

- **Registered Name:** Kagiso Mechanical Engineering (Pty) Limited
- **Trading As:** Kagiso Mechanical Engineering
- **Business Premises:** Rustenburg Industrial Zone, Rustenburg, North West Province
- **Client Reference:** CLI-066
- **Assigned IPv4 Supernet:** 192.168.35.0 / 255.255.255.0

2.2 Nature of Business

Kagiso Mechanical Engineering operates as a turnkey mechanical support partner to the mining houses and industrial clusters of the Bushveld Complex. Its core service offerings include:

- Heavy-duty mechanical component fabrication (boiler-making, structural steelwork)
- Planned and breakdown maintenance on mining processing plant equipment
- Reverse engineering and reclamation of wearing parts
- Non-destructive testing (NDT) and quality assurance inspections
- Project engineering and construction management services
- Hydraulic and pneumatic system design and installations

The company's clientele includes tier-one platinum producers, chrome beneficiation plants, and cement manufacturers located within a 150-kilometre radius of Rustenburg.

3. COMMERCIAL IMPERATIVES & STRATEGIC ALIGNMENT

Every technical decision made during this network redesign must demonstrate a clear line of sight to the following business drivers:

1. **Reduction of Production Downtime:** The network must underpin, and never interrupt, the workshop fabrication schedule. Unplanned outages directly translate to missed delivery deadlines and contractual penalties.
 2. **Protection of Intellectual Property:** Kagiso's design library of custom engineered components represents accumulated know-how spanning 18 years. This asset class requires stringent access control.
 3. **Scalability for the Bakwena Mine Expansion:** A recently awarded three-year maintenance contract at the Bakwena platinum shaft is expected to increase the engineering headcount by 30% within 24 months. The network must absorb this growth without forklift upgrades.
 4. **Compliance with Mining Charter B-BBEE Audits:** Financial records, HR files, and vendor contracts must be stored in logically segregated repositories and access must be auditable.
 5. **Engineer Productivity Uplift:** Draughtsmen currently spend up to 45 minutes per day waiting for large SolidWorks assemblies to replicate across the network. A 5x improvement in file transfer throughput is the target.
 6. **IT Support Optimisation:** Network faults must be diagnosable within 15 minutes. A flat, unsegmented network currently makes fault isolation a labour-intensive process.
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4. DEPARTMENTAL BREAKDOWN AND USER POPULATIONS

4.1 Staff Complement and Device Inventory Projection

Cost Centre	Functional Role Profile	Headcount	Networked Devices
Executive & Admin	MD, Financial Manager, HR Officer, Reception, PA x 2	8	10 PCs + 1 MFP
Engineering Design	Senior Engineers x 3, Design Draughtsmen x 8, BTech Interns x 4	15	15 CAD Workstations + 3 Plotters + 2 Shared PCs
Production Workshop	Foremen x 2, Artisans x 10, Machine Operators x 8	20	5 Kiosk Terminals + 5 Tablets + 2 Label Printers + 3 Workshop PCs
Finance & Procurement	Financial Accountant, Creditors Clerk, Costing Clerk, Buyer x 3	6	6 PCs + 1 Secure Printer + 1 Scanner
ICT & Technical Support	Network Admin, Systems Admin, Field Support Tech	3	3 Admin Laptops + 1 NAS + 1 Monitoring WS
Grand Total		52	~58 Devices

Cross-reference with D2 Physical Topology BOM: Milestone 1 Packet Tracer canvas physically draws **25 representative devices** (13 infrastructure + 4 departmental printers + 4 department PC-01 left flank + 4 department PC-02 below flank) — see Deliverable 02 Figure 1. The remaining **33 full-fleet devices** (per headcount projection above) will be bulk-added at Milestone 2 implementation; their 25-device skeleton placements on the 4 ALS access switches use ports Fa0/1 (PC left), Fa0/2 (PC below), Fa0/3 (Printer right), with ports Fa0/4 onwards pre-designated for future fleet roll-in.

4.2 Department-Specific Functional Requirements

Executive, Administration & Human Resources - Terminal access to the Sage VIP payroll platform (role-based, read-only access for HR Officer) - Confidential personnel file storage on the file server with folder-level permissions - MS Office 365 productivity suite and email - Shared access to two multi-function printers (colour for marketing, monochrome for admin) - Visitor management kiosk in the reception area

Engineering Design Office - Priority bandwidth for SolidWorks, AutoCAD, and Autodesk Inventor model repositories - Synchronisation with the central drawing office vault (minimum 1Gbps dedicated file transfer lane) - Access to the ISO 9001 quality document repository - Plotting capabilities for A0 and A1 drawing sets - Restricted read-only access to finance for quotation costing purposes only

Production Workshop Floor - Hardened kiosk terminals for work-order pull-sheet access - Machine-monitoring dashboard feeds from the CNC lathes and milling centres - Label printing for work-in-progress tracking - Strictly whitelisted internet access (only OEM parts portals and MSDS chemical safety sheets permitted) - No access whatsoever to the finance or HR network segments

Finance & Procurement Division - Highest-tier security classification on the LAN - Exclusive access to the Sage X3 ERP financial module and creditors ledger - EFT approval workflow terminal (air-gapped equivalent through ACL enforcement) - Procurement team access to vendor RFQ portals - Weekly automated backup of all financial records to the on-site NAS and off-site cloud storage

ICT & Technical Support Division - Full privileged access to every network segment for troubleshooting purposes - SSH / terminal access to all managed switches, routers, and server consoles - RADIUS / TACACS+ (simulated) authentication for all configuration changes - Network monitoring dashboard access (SNMP polling) - Deployment of WSUS and endpoint protection definition updates

5. EXISTING NETWORK SHORTFALLS AND MOTIVATION FOR REDESIGN

5.1 Diagnosis of Current Network Woes

The incumbent network at Kagiso Mechanical Engineering can be classified as a **first-generation flat peer-to-peer LAN**. It was cobbled together incrementally over a decade by various IT service providers without a master architecture. The consequences of this organic (and ungoverned) growth are:

1. **Broadcast Storm Saturation:** Every device sits in the same 192.168.1.0/24 broadcast domain. Wireshark captures by the part-time IT consultant routinely show broadcast traffic exceeding 18% of utilisation during morning logon, with ARP storms of 2–3 minute duration occurring at least once per fortnight.
2. **File Transfer Gridlock:** A 350MB SolidWorks assembly taking 11 minutes to copy across the network is the norm, not the exception. The design team's weekly drawing release procedure has evolved into an after-hours task simply to avoid the throughput collapse.
3. **Zero Information Boundaries:** An artisan in the workshop can, without any technical impediment, browse to the \Finance\Salaries share simply by guessing the server name. This represents a material governance failure under the POPI Act (Protection of Personal Information Act 4 of 2013).
4. **IP Address Sprawl and Duplication:** Static IPs assigned ad-hoc, combined with a rogue DHCP server on the old Netgear switch in the workshop, create a state where IP conflicts are logged in the IT register on average 4–6 times per month.
5. **No Change Control Regime:** The absence of any network documentation means a faulty NIC causing broadcast storms requires a process-of-elimination hunt that can last up to 6 hours. This is operationally unacceptable for a workshop that runs a 21-hour production day.

5.2 Case for Intervention

The redesign is justified on three pillars: governance (POPIA and audit compliance), productivity (engineer throughput), and resilience (fault isolation). This specification document translates those three pillars into 37 discrete, testable technical requirements.

6. TARGET NETWORK ARCHITECTURE: FUNCTIONAL REQUIREMENTS

6.1 Physical Topology Mandate

The design must instantiate a classic **three-tier Cisco hierarchical model**:

- **Core Tier:** Single ISR-series router (or equivalent Packet Tracer device) providing default gateway services, access-list enforcement, and simulated internet breakout. The core router is the trust boundary for all external traffic.
- **Distribution Tier:** One Layer-3 capable switch (3560 / 3650 class) acting as the inter-VLAN routing boundary and distribution point for all access-layer closets. Router-on-a-stick is acceptable as a Packet Tracer simplification, with a clear note in the design rationale that production deployment would leverage Layer-3 switching.
- **Access Tier:** One access switch per departmental zone. All end-user devices, printers, kiosks, and access points land here. Access ports are configured for a single data VLAN (no multi-VLAN trunking to end stations).

Physically, the core router and distribution switch reside in the dedicated IT server closet on the first floor. Vertical risers run to the second-floor design office and to the workshop sub-closet, each terminated on a 24-port access switch. Cat-6 copper is the specified medium; fibre is recommended for the 95-metre riser run to the workshop but Cat-6 is acceptable for the Packet Tracer simulation.

6.2 Logical Topology Mandate

The network must implement the following VLAN schema:

VLAN ID	VLAN Descriptor	Associated Subnet (CIDR + Mask, filled D4 VLSM)	Default Gateway (DLS1 SVI, filled D4 Section 2.1)
10	VLAN_ADMIN_HR	192.168.35.32/28 (255.255.255.240)	192.168.35.33
20	VLAN_ENGINEERING_DESIGN	192.168.35.0/27 (255.255.255.224)	192.168.35.1
30	VLAN_PRODUCTION_FLOOR	192.168.35.48/28 (255.255.255.240)	192.168.35.49
40	VLAN_FINANCE_PROCURE	192.168.35.96/28 (255.255.255.240)	192.168.35.97
50	VLAN_ICT_TECHNICAL	192.168.35.64/28 (255.255.255.240)	192.168.35.65
99	VLAN_MGMT_SERVERS	192.168.35.112/28 (255.255.255.240)	192.168.35.113
100	VLAN_GUEST_WIFI	192.168.35.80/28 (255.255.255.240)	192.168.35.81

Inter-VLAN communication is **denied by default**; explicit ACL entries will permit only the authorised traffic flows enumerated in the security section (Section 9).

6.3 Routing Philosophy

- **Intra-Site Routing:** Static routes are preferred for a network of this size and Packet Tracer fidelity. Alternatively, a single-area OSPF domain (Area 0) may be configured between the core router and distribution switch for academic demonstration purposes.
- **Default Route:** A single 0.0.0.0/0 default route on the core router points to the simulated ISP next-hop.
- **NAT:** Port Address Translation (PAT / NAT overload) is configured on the external-facing router interface so that all internal RFC-1918 addresses are translated for internet connectivity.

6.4 Equipment Schedule (Packet Tracer BOM)

Equipment Item	Qty	Justification
Cisco ISR 4331 (or 2911) Router	1	Core routing, NAT, ACL boundary, internet edge
Cisco Catalyst 3560-24PS Switch	1	Distribution layer, Inter-VLAN routing, PoE for APs
Cisco Catalyst 2960-24TT Switch	1	Admin / HR department access closet
Cisco Catalyst 2960-24TT Switch	1	Engineering Design Office access closet
Cisco Catalyst 2960-24TT Switch	1	Production Workshop Floor access closet
Cisco Catalyst 2960-24TT Switch	1	Finance Department (segregated closet, fewer ports — see substitution note below)
Cisco Aironet 2602 Access Point	2	One in admin wing, one in design office (PoE powered)
Generic Server (Server-PT)	2	DC/DHCP/DNS + File/Print services
Generic End Devices (PC-PT, Laptop)	58	Breakdown: 25 devices drawn on M1 canvas (13 infrastructure + 12 skeleton end-user: 8 PCs + 4 departmental printers) + 33 bulk fleet devices rolled in at M2 (per Section 4.1 headcount table)
Multi-Function Printers / Plotters	5	4 departmental printers drawn on M1 canvas (1 per ALS: ADM/ENG/PRD/FIN) + 1 plotter/MFP future roll-in per Engineering design office headcount
Copper Straight-Through Cables	~75	Horizontal cable runs
Copper Cross-Over / Crossover	4	Switch-to-switch uplinks (if Auto-MDIX not relied upon)

Finance Access Switch Substitution Note (BOM row 6): The brief originally listed a **Catalyst 2960-8TC** as the 8-port Finance dedicated switch. This specific model no longer appears in the Cisco Packet Tracer 9.0.1 default device catalogue (checked on 25 August — searching for **2960-8TC** returns zero results). It was swapped out for the **Catalyst 2960-24TT** instead, which still delivers on every design intent: 1. Finance retains its **own dedicated physical switch** inside the locked Finance Suite (not the general comms room). The 16 unused ports on the 24-port model sit empty — no non-Finance devices are ever plugged into them, so the physical separation guarantee is preserved. 2. It mounts in the same locked wall-cabinet originally planned for the 8TC, so physical key-lock access stays with the Finance Manager only. 3. Internally it runs the same 16 Gbps backplane, same MAC table size, same VLAN/trunking support, so performance or configuration at M2 is unaffected. 4. To confirm the 8TC is actually missing from Packet Tracer 9.0.1: open the device search bar, type **2960-8TC**, hit Enter — 0 devices are returned.

7. PERFORMANCE BENCHMARKS AND TECHNICAL STANDARDS

7.1 Throughput and Latency Commitments

Metric	Mandated Floor
Switch backplane inter-connectivity (Core-Dist)	≥ 1 Gbps LACP EtherChannel (2x GbE bundled)
Distribution-to-Access uplink capacity	≥ 1 Gbps (single or bundled)
End-station drop speed	100 Mbps min / 1 Gbps target
Access Point radio standard	802.11n (HT20/40) or 802.11ac
Peer-to-peer file transfer within Engineering VLAN	Transfer 350 MB in ≤ 90 seconds (equates to ~31 Mbps effective throughput)
End-to-end intra-site round-trip time (ICMP echo)	Mean ≤ 8 ms; 95th percentile ≤ 15 ms
Packet loss across the core	≤ 0.1% during steady-state operation

7.2 Congestion Management and Class of Service

The following DSCP / CoS markings shall be applied on egress at the distribution layer:

Class	DSCP	Traffic Profile	Queue Weight
NC	CS6	Routing protocols (OSPF hello, EIGRP hellos), Spanning-Tree BPDUs	High PQ
P1	EF	(Future) VoIP and emergency radio-over-IP	25% BW
P2	AF41	ERP transactions (Sage X3), drawing vault replication	30% BW
P3	AF21	General file share, email, print spooling	30% BW
BE	DF	Casual browsing, YouTube (only on guest VLAN 100)	15% BW

8. INFRASTRUCTURE SERVICES CATALOGUE

8.1 Dynamic Host Configuration Protocol (DHCP)

- Each VLAN corresponds to a unique DHCP scope.
- A central Windows Server (simulated in Packet Tracer) acts as the DHCP authority.
- **IP helper / DHCP relay** is enabled on every VLAN SVI (or router subinterface) so that unicast DHCP requests reach the central server.
- DHCP lease durations:
- Wired desktop devices in static seating areas: **7 days**
- Wireless clients and workshop tablets: **8 hours**
- Network test bench and training lab: **1 hour**
- Excluded ranges at the bottom of each scope are reserved for static assignments: servers, printers, managed switches, router SVIs, access points, and kiosk terminals.

8.2 Domain Name System (DNS)

- An internal DNS namespace of `kagiso-mech.local` is authoritative for the site.
- Forward lookup zones for all internal servers and infrastructure:
- `dc01.kagiso-mech.local` → DNS/DHCP/AD server
- `fs01.kagiso-mech.local` → File & Print server
- `sw-dist-01.kagiso-mech.local` → Distribution switch management address
- `rtr-core-01.kagiso-mech.local` → Core router management address
- Conditional forwarding for all queries outside the `kagiso-mech.local` zone is forwarded to Google public DNS (8.8.8.8 / 8.8.4.4) or the simulated ISP DNS in Packet Tracer.

8.3 File & Print Services

- A distributed share structure on FS01:
- `\\fs01.kagiso-mech.local\Engineering` → Engineering VLAN full control; Admin read-only; Finance read for costings; Production denied
- `\\fs01.kagiso-mech.local\Finance` → Finance VLAN full control; ICT admin full access; all others explicitly denied
- `\\fs01.kagiso-mech.local\HR` → HR and Directors full access; ICT admin full access for backup; all others denied

- `\\fs01.kagiso-mech.local\Shared` → All internal users read/write; Guest VLAN denied
- `\\fs01.kagiso-mech.local\Workshop_Instructions` → Production read-only; Engineering full control; others denied
- Network printers are published in Active Directory and pushed by GPO (conceptual; simulated in Packet Tracer via direct IP printing configuration).

8.4 Internet Breakout and Edge Services

- All internet access traverses a single NAT boundary at the core router (single-homed design).
- A Cisco reflexive ACL or zone-based firewall (Packet Tracer permitting) provides basic stateful inspection.
- Site-to-site VPN is out of scope for this phase.
- NTP time synchronisation from the core router to all internal infrastructure.

9. INFORMATION SECURITY & ACCESS GOVERNANCE

9.1 Inter-VLAN Traffic Matrix

The default security posture is **DENY ALL**. Only the flows enumerated below are explicitly permitted:

Source VLAN	Destination VLAN	Permitted Services / Justification
VLAN 10 (Admin/HR)	VLAN 99 (Servers)	SMB/CIFS (file shares), DNS, DHCP, RDP (to RDS only), Email
VLAN 10 (Admin/HR)	VLAN 20 (Eng)	SMTP (email only) – HR sends employee change notices to engineering supervisors
VLAN 20 (Eng)	VLAN 99 (Servers)	SMB, DNS, DHCP, NTP, HTTP/HTTPS to drawing vault
VLAN 20 (Eng)	VLAN 40 (Finance)	TCP 80/443 (quoting web portal only; no file share browse)
VLAN 20 (Eng)	VLAN 30 (Prod)	SMB (Workshop_Instructions read-only share), DNS
VLAN 30 (Prod)	VLAN 99 (Servers)	HTTP(S) OEM parts portals (whitelisted URLs), DNS, DHCP, NTP
VLAN 30 (Prod)	VLAN 40 (Finance)	DENY – ABSOLUTELY NO ACCESS PERMITTED
VLAN 30 (Prod)	VLAN 10 (Admin)	DENY – ABSOLUTELY NO ACCESS PERMITTED
VLAN 40 (Finance)	VLAN 99 (Servers)	SMB, DNS, DHCP, HTTPS (ERP), FTP (backup repository)
VLAN 40 (Finance)	VLAN 20 (Eng)	SMTP (purchase order notifications); ICMP echo unreachable
VLAN 50 (ICT)	ALL VLANs	SSH, Telnet (discouraged, permitted only from management PC), ICMP, SNMP – for operations support
VLAN 99 (Servers)	ALL VLANs	Response traffic only (established/related via reflexive ACL or context-based access control)
VLAN 100 (Guest)	Internet Only	HTTP/S, DNS forwarded. ALL internal RFC1918 destinations EXPLICITLY DENIED.
ALL Internal VLANs	Internet	PAT translated HTTP/S, SMTP for relay, IPSEC (future); FTP outbound restricted via proxy

9.2 Perimeter and Device Hardening Checklist

- **Switch Port Security** on every access port: maximum 2 MAC addresses (one PC, one IP phone/AP emulated), violation mode `restrict`, sticky MAC learning where practical.

- **VLAN Hopping Mitigation:** All switch-to-user ports set with `switchport mode access` and `switchport nonegotiate`. DTP disabled globally.
- **Private VLAN Edge (Protected Ports):** Guest WiFi VLAN 100 ports are isolated from each other.
- **Router Hardening:**
 - `service password-encryption` and `enable secret` enforced
 - Console line: password + login
 - VTY lines 0–15: ACL 10 restricted to VLAN 50 (ICT) only; `transport input ssh`; Telnet disabled
 - MOTD banner: "UNAUTHORISED ACCESS TO THIS SYSTEM IS PROHIBITED. ACTIVITY IS LOGGED AND MONITORED."
- **Spanning-Tree Best Practice:**
 - BPDU Guard enabled on all access ports
 - Root Guard on all ports facing downstream access closets where a rogue switch could be plugged in
 - Explicit root primary election on distribution switch

10. IPv4 ADDRESSING BOUNDARIES AND CONSTRAINTS

10.1 Delegated Address Space

The North-West University module coordinator has formally allocated the following private IPv4 prefix for the exclusive use of Client CLI-066 / Kagiso Mechanical Engineering:

Base Network: 192.168.35.0 / 24 **Subnet Mask:** 255.255.255.0 **Total Hosts Available (Pre-Subnetting):** 254 usable addresses (192.168.35.1 through 192.168.35.254)

10.2 Subnetting Strategy Rationale

The IPv4 subnetting design must simultaneously satisfy four (4) constraints: 1. **Sufficiency:** Every department has at least as many host addresses as its peak device count, multiplied by a 1.6x headroom factor. 2. **Efficiency:** VLSM (Variable Length Subnet Masking) is employed so that a 6-host department does not waste a /24 sized allocation. 3. **Growth:** The Finance department (currently 8 devices) must accommodate the planned procurement controller headcount increase. 4. **Summarisability:** Where possible, subnets should aggregate on logical bit boundaries to keep future route tables compact.

The final IP addressing plan will be submitted as a separate companion document (04_IP_Addressing_Plan.md) and will include: - Subnet ID, broadcast address, and first/last usable host for every VLAN - Static IP reservation table (routers, switches, servers, printers, APs) - DHCP pool start and end addresses plus exclusions - Point-to-point subnet assignments (if router-to-router links are used) - Change-control procedure for any future IP modifications

10.3 Address Assignment Policy

Device Class	Assignment Method
Router SVIs / Subinterfaces	Static – low end of subnet
Switch Management IP	Static – low end of subnet
Domain Controllers, File Servers	Static – fixed reservations
Network Printers / Plotters	Static – DHCP excluded range
Wireless Access Points	Static – DHCP excluded range
Kiosk Terminals (Workshop)	Static (for audit trail purposes)
Desktop PCs (Admin, Eng, Finance)	Dynamic DHCP – MAC reservations for finance if required
Laptops / Wireless Clients	Dynamic DHCP (short lease)
Visitor / Contractor WiFi	Dynamic DHCP from Guest VLAN pool

10.4 Non-Negotiable Change Control

- No employee may self-assign a static IP address under any circumstance.
 - All address allocations must be recorded in the central IPAM spreadsheet maintained by the ICT department.
 - Any subnetting change requires a signed CAB (Change Advisory Board) minute from the Financial Manager and MD.
 - **CRITICAL ACADEMIC CONSTRAINT:** The solution submitted for marking MUST remain 100% within the 192.168.35.0/24 block. Borrowing addresses from adjacent network ranges (e.g., 192.168.34.0/24 or 192.168.36.0/24) will be treated as a non-submission for the IP criterion.
 - **CRITICAL ACADEMIC CONSTRAINT:** This network design MUST remain strictly bound to Client CLI-066 and Kagiso Mechanical Engineering. No scenario substitution is permitted.
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11. IMPLEMENTATION GUARDRAILS AND COMPULSORY STANDARDS

The following hard constraints are drawn from the CMPG 325 Project Handbook and are non-negotiable:

1. **Packet Tracer Fidelity:** Every design choice must be fully realisable within the current Cisco Packet Tracer build. No reference may be made to appliances (Fortinet firewalls, F5 load balancers, Aruba controllers) that do not ship with Packet Tracer.
 2. **Academic Honesty (Turnitin):** All submission artefacts must be the student's original work. Where reference to external industry design guides is made, these must be cited using Harvard NWU referencing style.
 3. **CMPG 325 Handbook Alignment:** All configuration must be consistent with the prescribed module handbook's common project guidance. Where this specification introduces a department-specific requirement that appears to conflict with the handbook, the department-specific requirement prevails, and the rationale must be documented in the design justification memorandum.
 4. **Successful Data Exchange Demonstration:** The final Packet Tracer file must, at minimum, demonstrate a successful ICMP echo-request/reply between devices in different VLANs (as permitted by the ACL), and a successful file transfer (FTP or HTTP) from VLAN 20 (Engineering) to the file server on VLAN 99.
 5. **Assigned Networking Feature Integration:** In Milestone 2, the department's assigned networking challenge (whether DHCP, VLAN trunking, ACLs, EIGRP, OSPF, PAT, or another feature) must be correctly configured, validated, and documented with test screenshots.
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12. WORKING PRESUMPTIONS AND CAVEATS

The requirements in this document are compiled on the basis of the following working presumptions. If any presumption proves incorrect at implementation time, a Requirements Change Request must be logged and approved.

1. Kagiso Mechanical Engineering's Rustenburg facility is a single contiguous building; no multi-campus WAN overlay is required in Phase 1.
2. Electrical power and climate control for the IT server closet have been independently validated and are not within this project's scope.
3. The 192.168.35.0/24 delegation is the sole and sufficient public address block made available. No secondary private ranges will be routed.
4. A fibre-to-the-building internet connection (≥ 50 Mbps synchronous) is procurable from a Rustenburg-based ISP (e.g., Vodacom Business, MTN Business, or Frogfoot).
5. End-user PCs and CAD workstations will be refreshed within 6 months of network go-live to include Gigabit NICs. Older 10/100 Mbps legacy machines will be accommodated but will not achieve the throughput

benchmarks listed in Section 7.

6. A Windows Server 2019/2022 domain controller can be virtualised (2x VMs on a single physical host). For Packet Tracer purposes, two Server-PT devices will stand in.
 7. Printer fleet refresh (replacing the three ageing USB-connected HP LaserJets with network MFDs) is budgeted for Q4 2026. This specification assumes network-ready printers.
 8. The specific assigned networking feature to be implemented in Milestone 2 will be published per the module coordinator's schedule and will be integrated without requiring a full topology re-spin.
 9. POPIA compliance (Section 19, reasonable technical and organisational measures) is materially achieved by the VLAN and ACL controls in this document. A full POPIA audit is out of scope.
 10. All site civil works (wall boxes, trunking, cable laying, patch panel termination) will be subcontracted to a SACEPC-registered cabling contractor. This project covers design only.
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13. TERMINOLOGY REFERENCE SHEET

Abbreviation	Expanded Terminology and Contextual Definition
ACL	Access Control List – Ordered set of permit/deny statements evaluated against IP packet headers.
BPDU	Bridge Protocol Data Unit – Spanning Tree control frame exchanged between bridges.
CoS	Class of Service – Layer 2 (802.1p) 3-bit marking in an Ethernet frame header.
DHCP	Dynamic Host Configuration Protocol – RFC 2131 client/server protocol for automated IP leasing.
DNS	Domain Name System – Distributed hierarchical database mapping FQDNs to A/AAAA records.
DSCP	Differentiated Services Code Point – 6-bit field in the IPv4 ToS byte for packet classification.
DTP	Dynamic Trunking Protocol – Cisco-proprietary trunk auto-negotiation.
EIGRP	Enhanced Interior Gateway Routing Protocol – Cisco advanced distance-vector routing protocol.
FQDN	Fully Qualified Domain Name – e.g. fs01.kagiso-mech.local.
GPO	Group Policy Object – Active Directory container for policy deployment.
ICMP	Internet Control Message Protocol – Protocol used for ping and traceroute diagnostics.
ICT	Information and Communications Technology – Modern synonym for IT department.
ISR	Integrated Services Router – Cisco branch-router product family (e.g. ISR 4331).
LAN	Local Area Network – Single-site broadcast domain (or collection of VLANs).
LACP	Link Aggregation Control Protocol – IEEE 802.3ad standard for bundling multiple physical links.
MSDS	Material Safety Data Sheet – Chemical hazard information sheets required by OHS Act.
MFP	Multi-Function Printer – Combined printer, scanner, copier, fax device.
NDT	Non-Destructive Testing – Materials inspection methodology (ultrasonic, magnetic particle, etc.).
NAT	Network Address Translation – RFC 3022; specifically PAT overload for this design.
NTP	Network Time Protocol – RFC 5905 clock synchronisation protocol.
OSPF	Open Shortest Path First – Link-state IETF routing protocol (RFC 2328).
PAT	Port Address Translation – Many-to-one NAT (NAT overloading).
PC-PT	Packet Tracer PC – End-station device within the Cisco Packet Tracer simulator.
POPI Act	Protection of Personal Information Act 4 of 2013 – South African data privacy legislation.
PPTP / L2TP	Legacy VPN tunnelling protocols (not used in this design; mentioned for completeness).
RADIUS	Remote Authentication Dial-In User Service – Centralised AAA protocol.
RFC	Request for Comments – IETF published standards document.
SAGE X3 / VIP	Specific ERP and Payroll software packages used at Kagiso Mechanical Engineering.
SNMP	Simple Network Management Protocol – RFC 3410+ standard for device telemetry polling.

Abbreviation	Expanded Terminology and Contextual Definition
SSH	Secure Shell – RFC 4251 encrypted remote terminal protocol. Telnet is deprecated.
STP	Spanning-Tree Protocol – IEEE 802.1D loop-free active topology construction protocol.
SVI	Switched Virtual Interface – Layer-3 virtual interface on a Catalyst multilayer switch.
TACACS+	Terminal Access Controller Access-Control System Plus – Cisco AAA protocol (replaces TACACS).
VLAN	Virtual Local Area Network – IEEE 802.1Q tagged logical broadcast segmentation.
VLSM	Variable Length Subnet Masking – Allocation of different prefix lengths per subnet.
VPN	Virtual Private Network – Encrypted tunnel over an untrusted public transport network.
VTP	VLAN Trunking Protocol – Cisco-proprietary VLAN database propagation mechanism.
WSUS	Windows Server Update Services – Microsoft patch management platform.